TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	92.1 ± 0.30
Correlation Stability	87.8 ± 0.98
Distribution Consistency	94.8 ± 1.17
Entropy Preservation	95.9 ± 0.70
Mean Z Score Drift	0.0 ± 0.00
Categorical Fidelity	97.9 ± 1.04
Privacy Risk	1.2 ± 0.75
Data Density	100.0 ± 0.00
Coverage Score	76.1 ± 4.53
Feature Accuracy	94.8 ± 1.47
Novelty Score	0.9 ± 0.01

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Dataset Audit: indian_liver_patient_dataset.csv

1. Dataset Portrait

Attribute	Value
Rows	583
Columns	11
Numeric features	10
Categorical features	1
Missing cells (%)	0.1%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Age	44.746	44.391	-0.356	0.8%	OK
TB	3.299	3.291	-0.008	0.2%	OK
DB	1.486	1.445	-0.041	2.8%	OK
Alkphos	290.576	329.255	+38.679	13.3%	Warn
Sgpt	80.714	88.152	+7.439	9.2%	OK
Sgot	109.911	117.133	+7.222	6.6%	OK
TP	6.483	6.538	+0.055	0.8%	OK
ALB	3.142	3.171	+0.030	0.9%	OK
A/G Ratio	0.947	0.958	+0.011	1.2%	OK
Selector	1.286	1.264	-0.023	1.7%	OK

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	92.1 ± 0.30
Correlation Stability (*2)	87.8 ± 0.98
Distribution Consistency (*3)	94.8 ± 1.17
Entropy Preservation (*4)	95.9 ± 0.70
Mean Z Score Drift (*5)	0.0 ± 0.00
Categorical Fidelity (*6)	97.9 ± 1.04
Privacy Risk (*7)	1.2 ± 0.75
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	76.1 ± 4.53

Feature Accuracy (*10)	94.8 ± 1.47
Novelty Score (*11)	0.9 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7017094017094017	0.6735042735042736	0.7	-0.00170940170940171	
F1	0.8069728606563686	0.7772838960522617	0.8029868321714104	-0.00398602848495792	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7025641025641025	0.6529914529914534	0.6914529914529914	-0.01111111111111115	
F1	0.7997903571549948	0.7604696911120337	0.7886131990868901	-0.01117715806810486	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.5589743589743594	0.5034188034188034	0.5119658119658124	-0.04700854700854701	*
F1	0.5587432119380917	0.4721877026837605 4	0.4844697149805508 6	-0.07427349695754078	*

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6589743589743591	0.6367521367521368	0.6521367521367522	-0.00683760683760846	
F1	0.7622412399573748	0.7396895301222981	0.7639995774277231	0.00175833747034823 7	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6948717948717948	0.6581196581196582	0.6820512820512821	-0.01282051282051282	
F1	0.8037335702917616	0.765355036034334	0.7903219021091442	-0.013411668182617386	

Model: LR

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Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7059829059829059	0.6923076923076923	0.6957264957264957	-0.010256410256410265	
F1	0.8128002697439015	0.7996070740375031	0.8109500720906666	-0.0018501976532348085	

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6794871794871795	0.6752136752136753	0.6948717948717948	0.015384615384615385	
F1	0.7873306864277367	0.778636389516585	0.7953661816150248	0.008035495187288123	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6863247863247863	0.6598290598290598	0.6829059829059829	-0.0034188034188034118	
F1	0.7971299319768493	0.7788176460978614	0.7935140943975758	-0.0036158375792734176	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7068376068376068	0.7042735042735042	0.7068376068376068	0	
F1	0.8279091624529537	0.8256348655557642	0.8279091624529537	0	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6743589743589744	0.6777777777777778	0.6905982905982906	0.016239316239316227	
F1	0.7821343746683586	0.7767085573751971	0.7986687018321612	0.016534327163802643	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	65	Female	0.7	0.1	187	16	18	6.8	3.3	0.9
2	62	Male	10.9	5.5	699	64	100	7.5	3.2	0.74
3	62	Male	7.3	4.1	490	60	68	7.0	3.3	0.89
4	58	Male	1.0	0.4	182	14	20	6.8	3.4	1.0
5	72	Male	3.9	2.0	195	27	59	7.3	2.4	0.4
6	46	Male	1.8	0.7	208	19	14	7.6	4.4	1.3
7	26	Female	0.9	0.2	154	16	12	7.0	3.5	1.0
8	29	Female	0.9	0.3	202	14	11	6.7	3.6	1.1
9	17	Male	0.9	0.3	202	22	19	7.4	4.1	1.2
10	55	Male	0.7	0.2	290	53	58	6.8	3.4	1.0
11	57	Male	0.6	0.1	210	51	59	5.9	2.7	0.8
12	72	Male	2.7	1.3	260	31	56	7.4	3.0	0.6
13	64	Male	0.9	0.3	310	61	58	7.0	3.4	0.9
14	74	Female	1.1	0.4	214	22	30	8.1	4.1	1.0
15	61	Male	0.7	0.2	145	53	41	5.8	2.7	0.87
16	25	Male	0.6	0.1	183	91	53	5.5	2.3	0.7
17	38	Male	1.8	0.8	342	168	441	7.6	4.4	1.3
18	33	Male	1.6	0.5	165	15	23	7.3	3.5	0.92
19	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8
20	40	Female	0.9	0.3	293	232	245	6.8	3.1	0.8

B.1 Audit Trace: Fold 1 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	62	Male	0.9	0.2	140	16	21	6.3	3.8	1.85
2	40	Male	0.7	0.1	162	59	66	5.3	2.2	0.9
3	57	Female	0.8	0.1	176	28	22	6.7	3.0	0.5
4	22	Male	0.9	0.2	282	29	27	7.8	3.9	2.5
5	48	Female	0.8	0.2	298	33	29	5.9	0.9	0.47
6	38	Male	10.9	5.1	670	62	127	6.9	2.9	1.3
7	33	Male	7.1	3.7	196	622	497	6.9	3.6	1.09
8	38	Male	2.2	1.0	310	119	42	7.9	4.1	1.0
9	64	Male	3.0	1.4	248	46	40	6.5	3.2	0.9
10	40	Female	0.9	0.2	285	32	27	7.7	3.5	0.8

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	33	Male	5.8	3.3	182	15	63	6.1	3.2	1.9
2	22	Male	0.7	0.2	430	31	41	8.0	4.0	1.3
3	4	Female	2.5	1.6	192	42	27	6.8	3.6	0.8
4	35	Female	0.8	0.2	293	35	30	7.2	4.9	1.4
5	48	Male	0.8	0.1	298	91	145	6.0	2.1	0.95028077753779 7
6	49	Male	1.7	0.5	243	21	26	6.0	3.0	1.2
7	50	Female	1.0	0.3	191	22	31	7.8	4.0	1.0
8	52	Female	0.6	0.1	194	10	12	6.9	3.3	0.9
9	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
10	58	Male	0.8	0.2	298	33	59	6.2	3.1	1.0

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	17	Male	12.7	4.1	135	190	850	8.2	4.1	1.06
2	38	Male	1.1	0.7	220	61	142	6.3	3.9	0.4
3	14	Male	0.8	0.2	450	85	188	6.4	2.6	1.3
4	60	Male	0.4	0.2	216	12	200	7.4	3.3	0.8
5	45	Female	0.9	0.1	406	32	30	4.7	2.7	0.8
6	33	Male	1.0	0.3	168	22	42	7.0	3.8	1.5
7	44	Female	1.9	0.6	298	378	602	6.6	3.3	1.0
8	30	Male	0.8	0.2	182	46	57	7.8	4.3	1.2
9	53	Male	0.9	0.4	238	17	14	6.6	2.9	0.8
10	28	Male	0.8	0.3	190	20	14	4.1	2.4	1.4

B.4 Audit Trace: Fold 4 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	75	Male	2.9	1.4	388	30	21	6.6	2.8	1.0
2	60	Male	2.0	0.5	263	48	79	6.0	2.1	0.8
3	54	Male	0.8	0.2	154	32	42	8.0	4.2	1.3
4	45	Male	2.8	0.7	290	21	19	8.5	4.5	1.3
5	70	Male	0.8	0.2	234	16	24	7.2	3.2	0.8
6	36	Male	75.0	12.6	275	39	54	7.0	3.7	1.6
7	42	Male	0.7	0.2	197	64	33	5.8	2.4	0.7
8	69	Male	0.9	0.2	215	32	24	6.9	3.0	0.7
9	53	Male	1.6	0.9	178	44	59	6.5	3.9	1.5
10	49	Male	1.3	0.4	206	30	25	6.0	3.1	1.06

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	64	Male	0.7	0.2	310	31	55	7.2	2.4	0.5
2	18	Male	11.0	7.7	194	39	43	7.0	2.9	1.0
3	40	Male	2.7	3.0	320	22	20	7.1	4.3	1.4
4	35	Female	0.8	0.1	145	40	500	7.5	3.4	1.0
5	55	Male	2.3	3.0	719	29	22	6.1	3.2	0.6
6	49	Male	2.4	1.3	315	55	113	7.2	2.6	0.5
7	33	Male	2.1	0.7	205	50	38	6.8	3.0	0.7
8	22	Female	6.7	3.2	850	154	248	6.2	2.8	0.8
9	48	Male	5.8	2.5	802	133	88	6.0	2.8	0.8
10	84	Female	0.7	0.2	188	13	21	6.0	3.2	1.1

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	38	Male	0.7	0.2	192	26	24	7.4	4.9	0.5
2	72	Male	3.3	1.3	215	39	35	6.6	2.9	1.1
3	33	Female	0.8	0.2	198	19	40	6.0	2.2	0.6
4	35	Female	0.8	0.1	216	18	20	7.1	3.5	1.1
5	42	Male	3.0	1.2	268	29	63	6.7	3.1	1.0
6	51	Male	0.8	0.1	614	14	18	6.4	3.1	1.4
7	50	Male	0.9	0.3	194	190	73	7.5	3.9	1.0
8	49	Male	0.6	0.1	218	50	53	5.0	2.4	0.9
9	18	Male	0.8	0.2	228	55	54	6.9	4.0	1.3
10	30	Female	0.7	0.2	194	32	36	7.5	3.6	0.92

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	68	Male	0.8	0.2	257	390	66	5.6	3.0	0.64
2	62	Female	0.8	0.2	205	62	54	6.9	3.7	2.5
3	55	Male	8.6	7.0	350	74	25	6.9	3.6	0.9
4	65	Male	4.4	2.0	170	50	71	6.6	2.9	1.0
5	32	Male	1.8	0.4	190	1350	25	6.4	3.0	1.0
6	48	Female	1.3	0.6	224	69	731	5.1	1.5	0.4
7	13	Female	0.7	0.2	350	17	24	7.4	4.0	1.1
8	54	Female	23.2	12.6	574	43	47	7.2	3.5	0.9
9	63	Male	0.9	0.2	194	52	45	6.0	3.9	1.85
10	46	Male	0.6	0.2	290	26	21	6.0	3.0	1.0

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	32	Female	1.0	0.5	195	24	21	6.4	2.9	0.9
2	35	Male	0.8	0.2	298	29	24	5.0	2.0	0.4
3	58	Male	0.8	0.1	259	28	32	8.0	4.0	0.8
4	17	Male	0.9	0.2	187	58	126	5.9	3.1	0.8
5	23	Female	0.9	0.2	202	232	220	7.0	3.8	1.4
6	33	Female	15.8	4.1	194	25	16	7.5	4.0	1.0
7	48	Male	0.7	0.2	208	15	30	4.6	2.1	0.8
8	60	Male	22.8	12.6	962	53	41	6.9	3.3	0.9
9	27	Male	1.0	0.3	180	56	111	6.8	3.9	1.85
10	28	Female	0.8	0.2	309	55	23	6.8	4.1	1.51

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	40	Male	1.9	5.5	100	12	19	7.3	4.1	0.4
2	46	Male	0.7	0.1	208	102	143	7.1	4.3	0.9523160173160173
3	31	Female	0.8	0.2	205	29	67	4.6	2.0	1.1
4	62	Male	1.6	0.4	159	86	80	7.4	4.5	1.6
5	60	Female	0.8	0.2	490	56	40	6.8	3.2	0.7
6	32	Male	11.1	9.0	110	78	34	3.0	2.5	0.5
7	26	Male	2.0	0.9	195	24	65	7.8	4.3	1.2
8	47	Male	2.7	1.3	275	123	73	6.2	3.3	1.1
9	74	Male	1.0	0.3	175	30	32	6.4	3.4	1.1
10	48	Male	3.2	1.6	257	33	116	5.7	2.2	0.62

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Age	Gender	TB	DB	Alkphos	Sgpt	Sgot	TP	ALB	A/G Ratio
1	39	Male	3.5	3.3	195	25	87	6.1	3.0	1.2
2	42	Male	0.6	0.2	406	196	26	6.0	3.9	1.0
3	33	Male	0.6	0.2	298	32	18	6.4	2.7	0.8
4	23	Male	1.6	4.1	196	12	15	6.0	3.0	0.7
5	22	Male	0.6	0.2	195	26	21	6.8	2.9	1.6
6	38	Male	1.7	0.4	180	36	25	6.9	3.8	1.0
7	54	Female	5.5	3.2	350	67	42	7.0	3.2	0.8
8	43	Male	22.5	11.8	143	22	143	6.6	2.1	0.46
9	57	Male	4.5	2.3	315	120	105	7.0	4.0	1.3
10	65	Male	0.7	0.2	199	19	22	6.3	3.6	1.3